

LITTLE BLUE BOOK NO. 1477
Edited by E. Haldeman-Julius

How Glands Affect Personality

Grace Adams

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AFFECT PERSONALITY ***

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Grace Adams

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HOW GLANDS AFFECT PERSONALITY

WHAT IS PERSONALITY?

How do glands affect personality? Before any answer can be given to that question, before any discussion of the nature of glands can be undertaken, it is essential to agree upon a working definition of that currently popular but often rather ambiguous word—personality. What, precisely, does it mean?

Alluring advertisements in innumerable gaudy magazines promise, for a relatively small fee, to “develop your personality.” Even the pages of otherwise sedate and classical college catalogues boast of courses in “personality training.” And the term comes up incessantly in everyday conversation. We hear that a woman is beautiful but that she lacks personality; or that a man is not brilliant but that he has a great deal of personality. Such uses of the word undoubtedly fill a descriptive purpose. Most of us know what it means when it is employed in this manner. Yet it may be misleading. It implies that personality is a quality apart from physical and mental traits which some individuals possess to a high degree and in which others are totally deficient.

How can such an elusive attribute be either explained or defined? It cannot be, of course, but then it does not need to be for it is impossible to imagine a person without a personality. Every human being has, of necessity, a personality. It is personality which distinguishes him from some people and causes him to resemble others. And into its make-up goes every trait, physical, mental, emotional and temperamental, which he possesses; his sex, his height, his weight, the size and character of his features, his emotional stability, his mental powers, his special abilities. If these characteristics change to any great extent then his total personality is different.

Some of these traits, of course, seem more important, more ingrained in individuality than others. Sex, for example, appears more essential than

weight. A friend often becomes suddenly much stouter or thinner and yet retains the same character. On the other hand, the whole personality of the fat lady of the side show depends on her weight, that of the human skeleton on his weight and height combined, and of the bearded lady on her abnormal growth of hair. To a lesser degree such gross physical traits play an important role in the personalities of more normal individuals. And, speaking very generally, any marked anatomical variation is apt to be accompanied by finer mental and temperamental differences.

In everyday life we casually accept this correlation between physical and mental characteristics. We somehow unthinkingly divide the population into definite physiological types and we more or less expect these physiological types to show certain well marked traits of personality. In cartoons the reformer, the vice-crusader, is always portrayed as a tall, stiff, angular individual with a long, narrow face, a straight nose and tightly pressed lips. Without any accompanying label we would know that such a character was supposed to represent an intolerant, self-opinionated bigot. And without consulting our programs we can usually guess correctly which member of the cast of a musical comedy is the comedian. If there is an exceptionally plump person on the stage it is more than likely that it is he who will have most of the comical lines to repeat, who will be required to take the awkward tumbles and, in general, be responsible for the audience's laughter. For it is customary to expect fat people to be jolly and amusing. They look as though they should be and we feel disappointed if they turn out to be morose or dignified or too energetic. And in the same way we expect the sinister, dark mustachioed gentleman of the comic strips to be the villain, and the dainty, graceful maiden on the moving picture screen to be the sweet and guileless heroine.

These stock characters are, of course, highly conventionalized, but the convention is a natural, not an artificial one. It is natural to assume that certain physical types will show certain temperamental traits because usually they do. Often, however, they do not run true to form. And the personalities which surprise us by their anomalies are apt to be the ones which pique our curiosity, which arouse our interest and for whose individuality we seek an explanation. Yet it would be just as instructive to find a reason for those who conform to type as for those who differ; to learn why the thin, wiry, dark Frenchman is volatile, quick and romantic, while

the more stolidly built Scandinavian is slower, surer, and less excitable. We might, if it were not for the anomalies we have just noted, expect their respective physiques alone to account for their temperamental differences. But we would still have to explain the physiques, themselves. A great deal, of course, could be laid to inheritance. But what, exactly, is inherited? And why was the ancestor from whom the physiognomy was handed down built the way he was? And so we become involved in the whole question of racial differences, and it will be much safer to return once more to individual cases; to see if we can assign any reason not only for the fat man's plumpness but for his customary good humor, and for the reformer's thinness as well as his usual sourness of disposition.

For a long time such reasons were largely matters of speculation, but in recent years, particularly since 1889, science has turned its attention to this problem of personality. And its solution seems more and more to lie in an exact knowledge of the glands of internal secretion, or as they are often called, the endocrin glands.

OUR KNOWLEDGE OF THE ENDOCRIN GLANDS

The first physiologist to arouse any widespread interest among his fellow medical men in these small ductless glands was a Paris physician, Brown-Sequard. The experiments which he performed and the theories which he propounded have, for the most part, been disproved, but the enthusiasm which went into his investigations was soon caught by other scientists. In the last forty years masses of data have been collected which have thrown an increasingly brighter light on this subject which until comparatively recently was completely shrouded in darkness. Today candidates for medical degrees must have as full and up-to-date knowledge of the endocrin system as they have of the action of the heart and lungs and the disorders of digestion. Psychiatrists prescribe glandular treatments as supplements to their mental analyses. And attendants at feeble minded institutions and social workers must learn to recognize distinctive glandular types.

There have been two methods of gathering information about these glands whose importance for medicine and for psychiatry is becoming greater and greater: the clinical and the experimental. In the experimental method, which gives the more accurate results, animals are used as subjects. A certain gland is removed from a certain animal and then his growth, his movements, his whole life history subsequent to the operation is carefully noted and compared to the life histories of normal animals of the same species. Or the extract from another gland is injected into the blood stream of another animal and his behavior after the injection correlated with that of his fellows.

Human beings, of course, cannot be experimented upon in this ruthless fashion, but medical clinicians have found that nature, herself, has in many cases already done a good bit of experimenting. Some individuals are born with one or another of their endocrin glands either smaller or larger or more or less active than is customary. Several of these glands grow so close to the surface of the body that any abnormality of them can be easily seen. Others, which are more securely hidden, can be detected by the X-ray. And it has

been discovered that whenever an endocrin gland is exceptionally large or small or extremely active or noticeably sluggish, the individual in whom this abnormal condition occurs will exhibit other peculiarities, either of physical or mental development—most likely of both.

From these two sources, the clinic and the animal laboratory, some very definite information about extreme types of personality has been obtained, and the knowledge in many cases has been carried over to explain commoner but even more interesting variations of normal personality.

THE NATURE OF GLANDS

The human body contains many thousands of glandular structures, some large, some small, some simple, some complex, but all of them absolutely essential to its proper growth and health. Some of these organs which are most familiar are the sweat glands through the skin, the salivary glands of the mouth, the gastric glands of the stomach, the liver, the kidneys, the pancreas, and the male and female reproductive glands. Except for those of reproduction, the glands in this list are not to any especial extent connected with personality. Of course, if they do not function correctly, the health is impaired and changes in temperament result. But that much can be said of any other physiological organ. For the most part these glands of external secretion, as they are called, each has its own specific function to perform. They take up certain substances from the blood stream, or from the intestinal tract, transform them and then emit them through well defined ducts. In this manner the kidneys secrete urine, the liver bile, the gastric glands gastric juices. The ducts or tubes which carry the transformed substances to the exterior are essential parts of the glands, themselves, and the secretions are always ejected through them.

There are, however, other glandular structures in the body which have no definite outlets. The substances which they secrete do not pass out through well defined passages but are absorbed directly by the blood stream. Their effect is the same as that of drugs injected into the blood by a hypodermic needle. In fact, if the chemical constituent of a ductless gland's secretion is known and can be prepared in the laboratory, the same results can be got through its artificial injection as from the increased activity of the gland itself.

THE FUNCTION OF THE ENDOCRIN GLANDS

It is these ductless glands, or endocrin glands, or glands of internal secretion (they are known by any one of these names) which especially concern us when we speak of the effect of glands on personality. Besides the ovaries and testes, which give off an internal as well as an external secretion, they include the thyroid and parathyroid glands of the neck, the thymus gland in the chest, the adrenal glands situated just above the kidneys, and the pineal and pituitary bodies both located within the cranium close to the brain.

Because these organs have no visible outlets, it was only within comparatively recent times that they were supposed to have any function at all. Or if a purpose was assigned to them, it was usually highly fantastic and without any basis in fact. Thus the thyroids, which can be easily felt at the lower part of the neck on each side of the trachea, were supposed to keep the larynx moist and so produce a dulcet singing voice. But the thyroids, on account of their conspicuous position, are the glands about which we have the fullest information.

Because the secretions of the endocrin glands flow directly into the blood stream and are therefore carried very quickly to all parts of the body, their exact location is of slighter importance than might at first be thought. Because the pineal and pituitary bodies happen to be situated within the skull, they do not necessarily have a peculiar effect upon the brain. In fact, the thyroid in the neck probably is more closely connected with mental development than either of them. And the function of the pineal is more like that of the thymus than it is that of its nearer neighbor, the pituitary.

The thymus and the pineal. Broadly speaking, we may say that the thymus and the pineal are the glands which control childhood. Both are present and well developed at birth but continue to grow and expand for some years later. Each ceases to grow at puberty when the sexual glands become active. In normal individuals the pineal body disappears entirely

after adolescence and very little is left of the thymus subsequent to this period.

The internal secretion of the reproductive glands. Simultaneously with the atrophy of the thymus and the pineal body, the internal secretion of the reproductive glands, which up to this time have been dormant, commences and puberty sets in. The chief effect of the endocrin activity of the testes and ovaries lies in the development of the secondary sexual characteristics. In this connection the investigations of Steinach are the most authoritative as well as the most widely known. His experiments proved conclusively that it is not the reproductive elements proper but the interstitial cells of Leydig, which lie within the larger sex glands, which furnish the internal secretion. Professor Howell sums up Steinach's primary investigations as follows:

“Making use of very young animals (Steinach) has transplanted the testes from their normal position to other regions. Such animals develop normally, show all of the usual secondary sexual characteristics, and manifest full sexual desire and potency at the proper period. When the transplanted glands are examined the sexual elements are lacking, but the interstitial cells are increased in amount. It would appear from this work that sexual puberty is dependent upon the internal secretion furnished by these cells, and Steinach proposes to designate them collectively as the ‘puberty glands.’ This observer reports further remarkable experiments in which young males (rats, guinea pigs) were first castrated and then had transplanted under the skin or in the peritoneal cavity the ovary from a female of the same species. Under such conditions the graft of the ovary takes, and unlike the grafted testicle both the reproductive cells and the interstitial cells survive. In such animals the secondary male characteristics do not develop, his genital organs remain infantile; he exhibits, on the contrary, the female characteristics, as shown by his size, the character of the hair, and especially by the development of mammae and nipples. So far as the external characteristics are concerned the animal is completely feminized, and Steinach states that such an animal is sought by the male as though it were a true female.”

Working with very old animals, Steinach has further proved that by splitting the seminal duct, thus blocking the external secretion of the testes,

the interstitial cells can be made to function more vigorously and a temporary rejuvenation, a sort of second adolescence, can be produced.

Not all of the phenomena usually associated with puberty are, however, caused directly by the activity of the reproductive endocrine glands. Many of them are effected by the interstitial secretions stimulating to greater activity other ductless glands, namely, the pituitary, the adrenals and the thyroid.

The thyroids and parathyroids. So tightly are the parathyroids imbedded in the thyroids in all carnivorous and omnivorous animals, that many fatal operations were performed before it was discovered that the two are separate, if not completely independent organs. Goiters, which are always either enlargements of the thyroids, themselves, or of the connecting tissue surrounding the thyroids, can be easily eliminated by making a small incision in the neck and cutting away the superfluous tissue. In certain geographical regions where the soil contains very little iodine (the most important element in thyroid secretion) the thyroids, in an attempt to overcome this deficiency, grow abnormally large and endemic goiters result. In the country surrounding the Great Lakes of America and in the valleys of Switzerland such goiters are common. Often they are not large enough to be especially dangerous or conspicuous, but in some individuals they attain an enormous size and, of course, their possessors are anxious to have them reduce. Soon after the introduction of medical surgery, many beautiful Swiss ladies went into Paris to have their unbecoming tumors removed. And about half of those who were operated upon died within a few days. The rest recovered and showed no ill effects of their experience. Physicians were at a loss to account for the fatalities. But as animal experimentation was making headway, an explanation was soon forthcoming from the physiological laboratories.

The laboratory investigators discovered that if both thyroids were removed from carnivorous animals the subjects were within a few hours seized by severe convulsions and soon died. Herbivorous animals, on the other hand, lived for a long time after complete thyroidectomy, although they exhibited profound changes in what might be termed their personalities. Anatomical examinations showed that in all carnivorous and omnivorous animals the four tiny parathyroids are imbedded in the thyroid, while in herbivorous species two of the parathyroids are separated from and lie outside of the larger gland. From this it was decided that the thyroids,

although they are most important to proper development, are not essential to life, while the parathyroids, through the cancelling effect they have upon various poisons continually given off by the body, are. In the case of the unfortunate Swiss beauties, the parathyroids had been inadvertently cut away with parts of the thyroids in those who succumbed to the operation, but were left intact in those who survived.

The function of the thyroid. The thyroid is the endocrin gland about whose function we have the most complete information. Diseases of the thyroid, in particular goiters, are far from uncommon and the effects of these diseases have been carefully studied. Also the chemical constituents of the thyroid secretion has been determined and an artificial thyroid extract is now easily manufactured. This has been widely used in animal experimentation and as a human therapeutic. In normal individuals the thyroid glands function throughout life. In children they are necessary both to normal physical growth and to proper mental development. In adult life they are essential to body metabolism and mental stability.

The pituitary and the adrenals. Although the adrenal and the pituitary is each a single gland, both of them have two well defined parts. As in the case of the parathyroids and the thyroids, the anterior lobe of the pituitary and the cortex of the adrenals are essential to life, while the medulla of the adrenals and the posterior lobe of the pituitary furnish the secretions which have the more pronounced effect on the personality. Unlike the parathyroid, however, the life-essential portions of each of these glands is intimately connected with sexual development.

The chemical properties of the secretion produced by the medulla of the adrenal glands are known and form the basis of the drug called adrenalin. The effects of this drug have, of course, been carefully studied. The two most important results of an artificial injection of adrenalin lie in a strong stimulation of the nerves which control the heart, the blood vessels and the muscles; and an increase in the amount of sugar in the blood. Translated into more general terms, which will be explained in detail later, this means that the adrenals regulate our emotions as the thyroids do our mental life. And the posterior lobe of the pituitary body, in much the same manner, controls the muscles and skeletal bones directly.

From this brief outline of the role of the various ductless glands it should be evident that a well balanced person, a thoroughly normal individual, is one whose endocrin system is nicely attuned. His every gland secretes just enough but not too much. From the interstitial cells of his sexual glands he gets his secondary sexual characteristics, which until he reaches adolescence were kept in abeyance by his thymus and his pineal. His thyroid keeps his brain active; his pituitary body his muscles well toned; his adrenals his circulation well regulated and his emotional apparatus ready for emergencies. But, let us see what happens to an individual when one particular gland is either more or less active than it properly should be.

GLANDULAR ABNORMALITIES

The effects of heightened or lessened secretion of the sexual glands can be easily predicted. Abundant and very active interstitial cells cause the completely masculine man or the entirely feminine woman. When these cells are greatly reduced in number or dormant in their function, the sexual types are not so pronounced and approach each other in secondary characteristics. It is not until this condition becomes complicated by derangements of other ductless glands that it becomes definitely abnormal, but any variation in any one of these other organs is more than apt to have a direct effect on the reproductive system. This is shown clearly in the case of the thymus and the pineal.

The pineal gland, as we have seen, is essentially a gland of childhood. If both his pineal and his thymus are functioning normally then a child will be like other children. He will grow and develop gradually until he reaches his teens when the more drastic changes of puberty commence. It is at this time that his pineal body should cease to function and begin to atrophy. But suppose, for some reason, it begins to atrophy when he is only about six or seven years of age. What happens then? The child begins to grow remarkably, his reproductive organs develop, his figure becomes like that of an adult and he shows a strange mental precocity. This is the classical type of the child prodigy. And it was the pineal gland about which so much was said in the Leopold-Loeb murder trial.

If the pineal body continues to function well into the teens an individual who is almost the exact opposite of the one just described develops. Puberty is delayed for a long while—probably indefinitely. The child never grows up either anatomically, sexually or mentally.

The abnormal history of the thymus gland is similar to that of the pineal, but with a few slight differences. While the pineal and the thymus both act reciprocally with the sexual glands, they also have some check upon each other. It may be said in general that while both of these glands function in lieu of sexual secretion, the thymus allows the child to grow to its full stature while the pineal protects him from growing too rapidly. (It was to be

borne in mind, however, that both of these glands of childhood control growth indirectly, as we shall see when we come to study the pituitary in more detail.) Accordingly then, if a child has an insufficient thymus, his growth is stunted, his bones do not harden properly, he is subject to rickets and to malnutrition, he is apt to become either very fat or very thin; and there is a corresponding retardation of mental growth.

If the thymus continues to be active after the age at which puberty normally occurs, the results are more striking even than they are when this gland is insufficient during childhood. In this case the child proceeds to grow, but he grows only as a child. His reproductive organs do not develop and he acquires none of the characteristics which typify adolescence. Human beings of this class are not rare, but on account of the striking changes which occur when a tadpole matures, that is when its metamorphosis into a frog takes place, the result of a prolonged activity of the thymus is illustrated more clearly in the case of this animal than of any other.

Tadpoles fed upon a continual diet of thymus extract will not change into frogs. They remain tadpoles, but tadpoles of tremendous size. And a eunuch who has been castrated just before adolescence has, essentially, a tadpole personality. Because his interstitial cells never become active, no check is put upon his thymus. Instead of atrophying, it continues to enlarge. Consequently a eunuch is apt to be very tall, possibly also babyishly plump, but he undergoes none of the usual adolescent transformations. His voice remains high pitched, he does not grow a beard and all of his other characteristics remain almost completely childlike.

If thyroid extract is fed to young tadpoles an effect which is the exact opposite of that produced by thymus feeding results. The tadpoles do not grow, but metamorphosis into frogs takes place when they are still very tiny. Frogs scarcely larger than a fly can be developed in this manner. Thus it appears that not all of the processes usually associated with puberty are caused directly by the sexual glands, but rather by the internal secretions of these glands stimulating others to greater or lesser activity.

The thyroid, however, is essential to normal development in childhood and later life as well as at adolescence. In children an insufficient amount of thyroid secretion produces a state of arrested development known as

cretinism. Cretins are familiar figures in all feeble-minded institutions. They are small in stature, their skin is rough and coarse, their features are flat and characterless, and their tongues large and thick. They are awkward, unkempt and lazy. Mentally they are classified as idiots. Since the discovery of the active principle of the thyroid—thyroxin—cretinous children, if treated early enough, exhibit signs of marked improvement. They grow, they become more energetic and their mentality approaches more nearly to the normal.

When an insufficiency of thyroid extract occurs after adolescence, the disease known as myxedema sets in. Persons afflicted with this malady show definite and characteristic symptoms. Their hair begins to fall out, their skin becomes moist and swollen and they suffer continually from exhaustion. Most noticeable of all is a pronounced mental deterioration. This disease is commonly caused by an abnormal growth of the tissues surrounding the thyroid. This enlarged tissue crowds in upon the gland and causes it to shrink in size. It cannot under these conditions secrete the amount of extract necessary to bodily health, and myxedema results. The disease can be helped to some extent by regular doses of thyroxin and a permanent cure can be effected by an operation through which the superfluous tissue is removed and the thyroid allowed once more to resume its natural size.

An exophthalmic goiter produces results which are the exact opposite of those just described. In this case it is the thyroid gland, itself, which expands and therefore becomes hyperactive. The resulting excess in thyroid secretion causes the victim of the goiter to become extremely nervous and irritable (in contrast to the lassitude accompanying cretinism and myxedema). One of the most striking symptoms of this disease are the characteristic bulging, staring eyes. An operation by which the gland is cut down to its normal size alleviates the effects of the exophthalmic goiter.

When we were speaking of the growth of the skeleton, both in children and in adults, we noted that although it was influenced by the thymus, the pineal and the sexual glands, it was directly caused by another. This growth controlling organ is the pituitary. An excess of pituitary secretion in children brings about a condition known as gigantism. It is a hyperactive pituitary which is responsible for the giants to be seen in side shows, and a hypoactive one which occasions their fellow freaks, the midgets. An

insufficiency of the secretion of this gland causes pronounced obesity and an unnatural craving for sweets in adults.

The pituitary, like most of the other ductless glands, is also definitely related to the reproductive organs. If for any reason its development in a child is retarded, puberty does not occur. An individual showing eunuchoid tendencies results.

An increase in the amount of sugar in the blood and a superactivity of the entire circulatory system always follows an injection of adrenalin. This is given as proof of the fact that the adrenal glands have a definite control over the emotions. A brief summary of some recent investigations will explain this statement. Cannon of Harvard working with dogs, and Watson of Johns Hopkins using human subjects, discovered that the chief characteristics of the primary emotions, such as fear and rage, were an increase in blood pressure and an unusual supply of sugar in the blood. These two conditions constitute the physiology of an emotion, whatever its mental correlate may be. Cannon found further that the adrenals, as opposed to some of the other ductless glands, are richly supplied with nerve fibers. From this and from other experimental evidence he concluded that impulses from the brain stimulating the adrenals bring about the characteristic emotional symptoms. Of course these glands are not dormant at unemotional periods but under stress of mental excitement they become superactive. If the adrenal is unusually large and consequently congenitally hyperactive, an emotional state can be induced by a very mild stimulation. Consequently persons who possess exceptionally hard working adrenals can under very ordinary conditions become keyed up to a high pitch of excitement. Their hearts pound fiercely, their nerves are taut and they are constantly ready for action. Their digestions may suffer as a result of most of their energies being turned away from normal nutritive activities, but they can accomplish a great deal as long as their health continues.

Persons with a deficiency of adrenal secretion show opposite characteristics. They tire easily, they are continually depressed and they take very little interest in life.

Each gland which we have studied has been shown to have some connection with sexual development. The adrenals form no exception to this rule but their influence in this respect is even less straightforward than

that of the pituitary and thyroids. This is how Professor Berman describes the relation between the adrenal glands and sexual phenomena:

“In certain disturbances of these glands, especially when there are tumors, which supply a massive dose of the secretion to the blood presumably, peculiar sex phenomena and irregularities are produced. If the disease be present in the fetus, taking hold before birth, and so brought into the world with the child, there evolves the condition of pseudo-hermaphroditism. The individual, if a female, presents to a greater or less extent the external habits and character of the other sex. So that she is actually taken for a man, although the primary sex organs are ovaries, often not discovered to be such except when examined after an operation. How closely such an occurrence touches upon the problems of sex inversion and perversion comes at once to mind.

“If the processes involving the adrenal cortex attacks it after birth, the symmetrical correspondence and harmony of the secondary sex characters are not affected. But there follows a curious hastening of the ripening of body and mind summed up in the word puberty, a precocious puberty, with the most startling effects. A little girl of 2, 3, or 4 years of age perhaps will come to exhibit the growth and appearance of a girl of 14. She begins to menstruate, her breasts swell, she shoots up in height and weight, sprouts the hair distribution of the adult, and the mentality of the adolescent, restless, acquiring, doubting emerge. A tot bewitched into puberty! A boy of six or seven may suddenly, in the course of a few weeks or months, become a little man, robust, rather short and stocky, but moustached, with the muscular strength and sexual power of a man and thinking as a man....

“If the trouble in the adrenal cortex starts after puberty, phenomena of the same type, but of a different order, exhibit themselves. A woman, say in the thirties, becomes thus afflicted. Slowly or quickly her body will be covered by an abundant growth of hair, more or less of a beard and moustache appear upon the face, her voice will become deep and penetrating, her muscles will harden, and she will show a capacity for hard physical labor. Sexually she appears to be made over, masculinity now predominates in her make-up. Virilism is the name by which the French in particular have popularized the knowledge of the condition.”

GLANDS AND NORMAL PERSONALITY

Most of these cases which have been described, where one single gland is much too active or a great deal too sluggish, would be classified as extreme types of personality. They are, for the most part, definitely normal. Primarily because they are so distinctive it was fairly easy to ascertain just which form of glandular disorder conditioned which kind of mental or physical peculiarity. With personalities which conform more closely to the norm, the question of specific glandular control becomes more baffling. Yet when we note how accurately each specific glandular derangement is correlated with a definite peculiarity of personality, it seems highly probable that slighter variations in individuality may also be due to differences in the endocrin system. When we further realize that any one of the seven glands of internal secretion may be either a little overactive or a little underactive while the rest are normal, or that two or three or four or five glands may be slightly abnormal in one way or another, it begins to appear as though this complicated endocrin system may be responsible for all the lesser variations of personality as well as those which are more pronounced.

An individual whose thyroid is underactive will be different from an individual whose thyroid is overactive, and he will be still more unlike one whose pituitary and adrenals are also exceptionally forceful. Moreover X-ray pictures show that when any one gland is especially small and unable to grow larger in order to compensate for its deficiency, some other gland may expand and in some manner take over its function. So when we come to the study of normal personalities the role of the endocrin system may be much more complex than it is with the simpler and more distinctive glandular types. Our knowledge of the glandular control of the average man is still far from complete, but, remembering the striking symptoms of abnormal functioning, we can discern certain general glandular tendencies among mainly normal types. We cannot as yet say that Mr. Smith's thyroid is one hundred per cent efficient while his pituitary works only one half as energetically as it should and his adrenals twenty per cent actively. But we can say that from certain physical or mental traits which Mr. Smith shows

his whole personality is probably dominated by one or more of his endocrine glands.

If Mr. Smith is exceptionally tall and the bones in his face and hands and feet noticeably large, it is safe to assume that Mr. Smith has a very forceful pituitary body. Yet, if in spite of his height, Mr. Smith appears young for his age, has a smooth boyish face and a youthful outlook on life, then we may correctly infer that his thymus has been active longer than is customary with adults. Mrs. Smith, too, may seem much younger than her chronological age would lead one to expect, and she, also, may have a rudimentary thymus which is continuing to function. But if, as opposed to her husband, she is small and dainty, we would suppose that her pituitary is also smaller than his. And if with her doll-baby stature, she had a still childish mind and was not particularly energetic, we might further conclude that her thyroid was not accomplishing all that it should. If, as she grew older, she became suddenly nervous and irritable, it would seem evident that this gland was suddenly expanding in an attempt to overcome its earlier deficiencies.

Mr. and Mrs. Smith might have two children, a son and a daughter, who while they were young resembled each other rather closely. But as they grew older they might seem more and more different. Perhaps the girl could continue to grow rapidly after she was twelve years of age but would still appear childish. She would still play with her dolls and with younger children and take no interest in the activities which normally absorb the attention of a young girl. Then we would infer that the thymic tendency which she inherited from both parents was intensified in her case. If, however, her other glands were able to overcome this dominance of the thymus and so allow her to achieve adolescence, she might still never become so completely feminine as her mother. She would never be very interested in boys or young men, but would be enthusiastic about athletics and feminist movements. Then it would be reasonable to assume that her adrenal glands had taken over the supremacy once held by the thymus.

The Smith boy, on the other hand, might mature rapidly. He would become quiet and serious at the age when most boys are still playing marbles and hooky. His voice would change and his beard would begin to grow. In this case we could be certain that either his thymus or his pineal gland had deserted him too soon. If he stopped growing at this time it would

indicate that his thymus was failing him, but if he continued to become taller and his mind also matured quickly the evidence would be in favor of an atrophying pineal. If later in life he suddenly became exceedingly plump and rather listless, we might conclude that still another gland was not working as hard as it might. This time it would be his pituitary.

These cases are, of course, merely hypothetic and it is entirely too much to expect such a small family to show such divergent types and so many changes in personality. But it does indicate how glandular activity can affect the character and habits of persons who would not be considered abnormal. And taking the population as a whole, certain glandular tendencies do determine definite types of personality. The slow, stupid, colorless people have too little thyroid secretion. Those who are superactive mentally and temperamentally have a large amount of this extract. The very fat or very tall people have extraordinary pituitaries. The energetic, executive types are well supplied with adrenalin. And most of those who seem exceptionally youthful long after they have reached chronological maturity have had thymuses or pineals which have functioned longer than is natural in adult life.

Transcriber's Notes:

Punctuation and spelling inaccuracies were silently corrected.

Archaic and variable spelling has been preserved.

Variations in hyphenation and compound words have been preserved.

*** END OF THE PROJECT GUTENBERG EBOOK HOW GLANDS
AFFECT PERSONALITY ***

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